



Swiggy Restaurant Performance & Market Analysis

Business Problem

Background

Swiggy is one of India's leading food delivery platforms, connecting customers with restaurants across multiple cities. Restaurant performance on the platform can be evaluated using metrics such as customer ratings, pricing, cuisine type, delivery time, and geographic location.

Understanding these patterns can help identify operational opportunities, support restaurant performance, and improve customer experience.

Problem Statement

Swiggy aims to evaluate restaurant performance across major Indian cities by analyzing customer ratings, pricing patterns, cuisine categories, delivery efficiency, and customer engagement.

The objective is to assess restaurant and city-level performance, explore relationships between pricing, ratings, and delivery time, identify high-performing restaurants and markets, and generate data-driven

recommendations to support operational and customer experience improvements.

The analysis combines exploratory data analysis, advanced SQL, statistical analysis, and Power BI visualization to translate restaurant-level data into actionable business insights.

Business Objectives

1. Analyze the distribution of restaurants across cities and areas.
2. Identify cities with the highest restaurant concentration.
3. Evaluate customer rating patterns across restaurants, cities, and cuisine categories.
4. Analyze delivery-time performance across different locations.
5. Examine pricing patterns across cities and food categories.
6. Identify high-performing restaurants based on ratings and customer engagement.
7. Evaluate relationships between pricing, ratings, and delivery time.
8. Generate evidence-based recommendations to improve operational and customer experience outcomes.

Key Business Questions

- Which cities have the highest number of restaurants?
- Which cities have the highest average restaurant ratings?
- What are the most common food-type categories on the platform?
- How are restaurant ratings related to pricing?
- How are delivery times associated with customer ratings?
- How does delivery time vary across cities and areas?
- Which restaurants demonstrate consistently high ratings and customer engagement?
- Which markets and restaurant segments show notable performance patterns?

Expected Outcome

The analysis provides insights into restaurant performance, customer ratings, pricing patterns, delivery efficiency, and customer engagement. These findings can support operational improvements, restaurant-partner benchmarking, market analysis, and customer experience initiatives.

Project Analysis

Dataset Understanding

Methodology.

Data Quality Assessment

EDA(Exploratory Data Analysis).

Advanced Analysis

KPI Definition

Business Insights Summary.

Recommendations

Limitations

Power Bi Dashboard

Dataset Understanding

Dataset Name: Swiggy Restaurant Dataset

Source: (Kaggle)

<https://www.kaggle.com/datasets/abhijitdahatonde/swiggy-restuarant-dataset/data>

Total Records: Rows 8680, Columns : 10

Description:

The dataset contains information about restaurants listed on Swiggy including restaurant names, locations, pricing, ratings, cuisine types, and delivery times.

Columns:

Column Name	Data Type
ID	int
Area	nvarchar
City	nvarchar
Restaurant	nvarchar
Price	float
Avg_ratings	float
Total_ratings	smallint
Food_type	nvarchar
Address	nvarchar
Delivery_time	tinyint

Methodology

The analysis followed a structured workflow using SQL Server and Power BI:

1. **Business Understanding** — Defined the business problem, objectives, and key analytical questions.
2. **Data Understanding** — Reviewed the dataset structure, columns, data types, and business meaning of each field.
3. **Data Quality Assessment** — Checked for missing values, blank values, duplicate records, invalid prices, and data inconsistencies.
4. **Exploratory Data Analysis** — Used SQL to analyze restaurant distribution, city performance, cuisines, ratings, pricing, and delivery times.
5. **Advanced Analysis & KPI Development** — Applied CTEs, window functions, ranking, statistical measures, and business KPI calculations to identify important performance patterns.
6. **Power BI Dashboard Development** — Built two interactive dashboards to communicate executive-level and restaurant-level insights.
7. **Business Insights & Recommendations** — Interpreted the findings and translated them into practical business recommendations.

Data Quality Assessment

Total Rows = 8680

Data Types check:

SQL:

```
SELECT COLUMN_NAME,  
       DATA_TYPE  
FROM INFORMATION_SCHEMA.COLUMNS;
```

Column Name	Data Type
ID	int
Area	nvarchar
City	nvarchar
Restaurant	nvarchar
Price	float
Avg_ratings	float
Total_ratings	smallint
Food_type	nvarchar
Address	nvarchar
Delivery_time	tinyint

Finding: Correct

Null Value Analysis :

SQL CODE:

```

SELECT COUNT(*) - COUNT(ID) AS ID_Nulls,
       COUNT(*) - COUNT(Area) AS Area_Nulls,
       COUNT(*) - COUNT(City) AS City_Nulls,
       COUNT(*) - COUNT(Restaurant) AS Restaurant_Nulls,
       COUNT(*) - COUNT(Price) AS Price_Nulls,
       COUNT(*) - COUNT(Avg_ratings) AS Avg_ratings_Nulls,
       COUNT(*) - COUNT(Total_ratings) AS Total_ratings_Nulls,
       COUNT(*) - COUNT(Food_type) AS Food_type_Nulls,
       COUNT(*) - COUNT(Address) AS Address_Nulls,
       COUNT(*) - COUNT(Delivery_time) AS Delivery_time_Nulls
FROM swiggy;

```

Column Name	Value
ID	0
Area	0
City	0
Restaurant	0
Price	0
Avg_ratings	0
Total_ratings	0
Food_type	0
Address	0
Delivery_time	0

Decision: No action required.

Blank Values :

We Check Blank Values of these columns **Area, City, Restaurant, Food_type, Address**

```

SELECT
  SUM(CASE WHEN TRIM(Area) = '' THEN 1 ELSE 0 END) AS Area_Blanks,
  SUM(CASE WHEN TRIM(City) = '' THEN 1 ELSE 0 END) AS City_Blanks,
  SUM(CASE WHEN TRIM(Restaurant) = '' THEN 1 ELSE 0 END) AS Restaurant_Blanks,
  SUM(CASE WHEN TRIM(Food_type) = '' THEN 1 ELSE 0 END) AS Food_type_Blanks,
  SUM(CASE WHEN TRIM(Address) = '' THEN 1 ELSE 0 END) AS Address_Blanks
FROM swiggy;

```

Finding: Every column return 0

Decision: No action required.

Further analysis found that the Price column contained 0 values. Since a price cannot be 0, we investigated further.

SQL:

```
SELECT *  
FROM swiggy  
WHERE Price = 0;
```

Finding: 5 restaurant records contained a price value of 0

Affected Records:

ID

55365

55710

73662

77391

283144

Percentage of records with a price of 0 :

```
SELECT  
cast(round((count(*) * 100.0) /(SELECT count(*) FROM swiggy), 2) as decimal(10,2)) AS  
Percentage_Zero_Price  
FROM swiggy  
WHERE Price = 0;
```

Thats **0.06%** of the dataset. Investigation suggests these records likely originated from missing values in the source CSV that were converted to 0 during data import.

Decision:

Since a restaurant cannot realistically have a price of 0 and this represented only **0.06%** of the dataset and was not a valid business value. Price values of 0 were treated as missing values and excluded from price-based analysis.

Duplicate Check :

ID Check:

```
SELECT
    ID,
    COUNT(*)
FROM swiggy
GROUP BY ID
HAVING COUNT(*) > 1;
```

No duplicate Restaurant IDs found.

Decision: No action required.

Exact Duplicate Records Check:

```
SELECT
    Area,
    City,
    Restaurant,
    Price,
    Avg_ratings,
    Total_ratings,
    Food_type,
    Address,
    Delivery_time,
    COUNT(*)
FROM swiggy
GROUP BY
    Area,
    City,
    Restaurant,
    Price,
    Avg_ratings,
    Total_ratings,
    Food_type,
    Address,
    Delivery_time
HAVING COUNT(*) > 1;
```

Finding:

After checking, we found two rows with exactly the same values across all columns.

City = 'Ahmedabad', Area = 'Kuber Nagar', Restaurant = 'Cake Adda'.

Therefore, we decided to conduct further analysis before making a decision.

Further analysis:

```
SELECT *  
FROM swiggy  
WHERE ID IN(  
    SELECT ID  
    FROM swiggy  
    WHERE Restaurant = 'Cake Adda'  
    AND City = 'Ahmedabad'  
    AND Area = 'Kuber Nagar'  
);
```

Finding:

Identified 2 records (ID= 342814 and 343566) with identical business attributes (Cake Adda, Ahmedabad) but different IDs.

Decision: Since the dataset did not provide sufficient information to determine whether these represented true duplicates or separate listings, the records were retained.

EDA(Exploratory Data Analysis)

Q1: Which cities have the highest number of restaurants?

SQL-

```
SELECT    City,
          COUNT(*) AS Restaurant_Count
FROM      Swiggy
GROUP BY City
ORDER BY Restaurant_Count DESC;
```

Finding:

Kolkata leads with **1,346** restaurants — **9.7%** more than **Mumbai (1,227)** and **21.7%** more than **Chennai (1,106)**. Together, the **top 3** cities account for **3,679 restaurants** — **42.4%** of the entire platform's **8,680 listings**, showing high geographic concentration.

Business Interpretation:

Kolkata appears to be the most concentrated restaurant market within the dataset. The large number of restaurants suggests strong customer demand and high platform adoption. However, restaurant competition may also be higher compared to other cities.

Q2: Which cities have the highest average restaurant ratings?

SQL:

```
SELECT    City,
          COUNT(*) AS Restaurant_Count,
          ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM      Swiggy
GROUP BY City
ORDER BY Avg_Rating DESC;
```

Finding:

Chennai recorded the **highest average restaurant rating (3.78)** among major cities while maintaining a large restaurant base of over **1,100 restaurants**. This

suggests consistently strong customer satisfaction across a significant number of restaurants.

Additional Insight:

Kolkata

Restaurants = 1346 (Highest)

Rating = 3.70

- **Kolkata** has the **highest number of restaurants** in the dataset while maintaining an average rating of **3.70**, indicating a mature and competitive restaurant ecosystem.

Delhi

Rating = 3.53

- **Delhi's** average rating (**3.53**) sits **0.25** points below Chennai's platform-leading **3.78** — a **7%** satisfaction gap. Closing even half this gap (**0.12 points**) would bring Delhi closer to the platform average of **3.66**, potentially improving customer retention across its restaurant base.

Business Interpretation:

Chennai and Bangalore demonstrate stronger customer satisfaction levels compared to other cities in the dataset. Chennai's high rating combined with a large restaurant base suggests a healthy restaurant ecosystem capable of consistently meeting customer expectations.

Kolkata contains the highest number of restaurants while maintaining an average rating of 3.70, indicating strong market maturity and competition.

Delhi reported the lowest average rating (3.53), highlighting a potential opportunity for service quality improvements.

Q3: What are the most common cuisine categories on the platform?

SQL:

```
SELECT TOP 20 Food_type,
              COUNT(*) AS Restaurant_Count
FROM Swiggy
```

```
GROUP BY Food_type  
ORDER BY Restaurant_Count DESC;
```

Finding:

The top 5 cuisines — **Indian (389), Chinese (277), North Indian (246), Fast Food (240), South Indian (228)** — **account for 1,380 restaurants, representing 15.9%** of all platform listings. Indian cuisine alone holds **4.5%** of total platform share, making it the **single dominant** category by a **40.4%** margin over the **#2 cuisine (Chinese)**.

Business Interpretation:

Traditional Indian cuisines dominate the platform, indicating strong customer demand for familiar and locally preferred food options. Chinese and Fast Food categories also maintain a significant presence, reflecting growing demand for quick-service and international cuisine options.

The concentration of restaurants within a few major cuisine categories suggests that these segments represent the core food delivery market and should remain strategic priorities for restaurant acquisition and promotional activities.

Additional Insight:

top 5:

Indian
Chinese
North Indian
Fast Food
South Indian

These are all mass-market cuisines.

Insight:

The dominance of Indian, Chinese, and Fast Food categories indicates that customers primarily prefer familiar and frequently ordered cuisines rather than niche or premium food segments.

Q4: Do higher-priced restaurants receive better ratings?

SQL:

```
SELECT CASE WHEN Price < 200 THEN 'Low Price' WHEN Price BETWEEN 200 AND 400 THEN 'Medium Price' ELSE 'High Price' END AS Price_Category,
       COUNT(*) AS Restaurant_Count,
       ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM   Swiggy
WHERE  Price > 0
GROUP BY CASE WHEN Price < 200 THEN 'Low Price' WHEN Price BETWEEN 200 AND 400 THEN 'Medium Price' ELSE 'High Price' END
ORDER BY Avg_Rating DESC;
```

Finding:

- **Higher-priced** restaurants have the **highest** average rating (**3.79**).
- followed by **low-priced** restaurants **2nd highest** average rating (**3.66**)
- Medium-priced restaurants have the **lowest** average rating (**3.61**).

Business Interpretation:

Customers appear to rate premium restaurants more favorably than budget or mid-priced restaurants. This may indicate that higher-priced restaurants provide better food quality, service, packaging, or overall customer experience.

Additional Insight:

High Price = 1787 restaurants

Medium Price = 5911 restaurants

Medium-priced restaurants dominate the platform.

Price tier distribution — the medium dominance story

SQL:

```
SELECT CASE WHEN Price < 200 THEN 'Low' WHEN Price <= 400 THEN 'Medium' ELSE 'High'
END AS tier,
       COUNT(*) AS cnt,
       ROUND(COUNT(*) * 100.0 / SUM(COUNT(*) OVER(), 1) AS pct,
       ROUND(AVG(Avg_ratings), 2) AS avg_rating
FROM   Swiggy
WHERE  Price > 0
GROUP BY CASE WHEN Price < 200 THEN 'Low' WHEN Price <= 400 THEN 'Medium' ELSE 'High'
END;
```



Medium-priced restaurants (₹200–400) dominate with **5,911** listings — **68.1%** of the platform. **High-priced restaurants (₹400+)** represent just **20.6% (1,787 listings)**, yet achieve the **highest average rating (3.79)**. This creates a clear quality-price signal: the premium segment outperforms the volume segment by **0.18** rating points despite being **3.3×** smaller in count.

Q5: How does delivery time vary across cities and areas?

- **Query 1: Average Delivery Time by City**

SQL:

```
SELECT    City,
          ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time
FROM      Swiggy
GROUP BY  City
ORDER BY  Avg_Delivery_Time;
```

Finding:

Ahmedabad recorded the **fastest** average delivery time (**44 minutes**), while **Kolkata** recorded the **slowest (67 minutes)**.

Insight:

Kolkata's average delivery time (67 min) is 52.3% longer than Ahmedabad's (44 min) — the largest city-level delivery gap on the platform. The platform **average** is **53 min**, meaning **Kolkata** runs **26.4% above average** while **Ahmedabad** runs **16.9% below it.**

- **Query 2: Delivery Time Distribution:**

SQL:

```
SELECT MIN(Delivery_time) AS Min_Time,
       MAX(Delivery_time) AS Max_Time,
```



```
ROUND(AVG(Delivery_time), 2) AS Avg_Time  
FROM Swiggy;
```

Finding:

The average delivery time across the platform is **53 minutes**. However, delivery times vary significantly from **20 minutes** to **109 minutes**, indicating substantial operational differences across restaurants and locations.

- **Query 3: Areas with Slowest Delivery**

SQL:

```
SELECT TOP 10 Area,  
              ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time  
FROM Swiggy  
GROUP BY Area  
ORDER BY Avg_Delivery_Time DESC;
```

Finding:

Several Kolkata localities consistently experience longer delivery times, with **Rabindrapally** averaging **97 minutes**. These areas may face operational challenges such as traffic congestion, delivery partner availability, or restaurant concentration issues.



Strong Business Insight

Earlier, we saw that **Kolkata** has the **highest restaurant count (1,346)**. However, after analyzing overall delivery performance and the areas with the longest delivery times, we found that Kolkata has the **slowest average delivery time**.

Business Interpretation:

Despite having the highest number of restaurants in the dataset, Kolkata also records the slowest average delivery time. This suggests that restaurant density alone does not guarantee faster deliveries and that operational factors such as traffic conditions, order volume, and delivery network efficiency may significantly impact performance.

Q6: Which restaurants consistently achieve high ratings?

Methodology

To identify consistently high-performing restaurants, only restaurants with **at least 100 customer ratings** were considered. This helps ensure that high ratings are supported by sufficient customer feedback and are not based on a small number of reviews.

SQL:

```
WITH RatedRestaurants
AS (SELECT Restaurant,
      Food_type,
      City,
      ROUND(Avg_ratings, 2) AS Avg_ratings,
      Total_ratings
   FROM Swiggy
  WHERE Total_ratings >= 100)
SELECT TOP 20 *
FROM RatedRestaurants
ORDER BY Avg_ratings DESC, Total_ratings DESC;
```

Business Insights:

1. High Ratings Can Be Sustained at Scale

Restaurants such as:

- Corner House Ice Cream (4.7)
- Mithai (4.7)
- New Kalpana Mistanna Bhandar (4.7)

maintain ratings of **4.7+** while receiving **1,000 customer ratings**, demonstrating consistent customer satisfaction at high order volumes.

2. Kolkata Has Multiple Top-Performing Restaurants

Several restaurants from Kolkata appear among the top-rated restaurants, including:

- Nic Natural Ice Creams (4.8)

- Mithai (4.7)
- New Kalpana Mistanna Bhandar (4.7)
- Mama Mia! - Italian Ice Creams (4.7)

Although Kolkata recorded longer delivery times in earlier analysis, the city also contains multiple highly rated restaurants, suggesting strong food quality and customer satisfaction.

3. Specialty category outperformance:

SQL:

```
SELECT  Food_type,
        COUNT(*) AS top_count
FROM (SELECT TOP 20 Food_type
      FROM      Swiggy
      WHERE     Total_ratings >= 100
      ORDER BY  Avg_ratings DESC, Total_ratings DESC) AS t
GROUP BY Food_type
ORDER BY top_count DESC;
```

Many top-performing restaurants specialize in:

- Ice Cream
- Desserts
- Bakery
- Coffee



Among the **top 20 highest-rated restaurants (min 100 reviews)**, specialty categories — **ice cream, dessert, bakery, coffee** — represent at least **12 of 20 entries (60%+)**, despite these categories being a small fraction of total platform listings. This suggests a **3–4×** higher probability of achieving top-tier ratings for specialty food brands vs mass-market cuisine restaurants.

Business Interpretation:

Restaurants such as **Nic Natural Ice Creams, Corner House Ice Cream, Mithai, New Kalpana Mistanna Bhandar, and Theobroma** emerge as the strongest performers on the platform. Their high ratings combined with substantial customer review volumes indicate consistent service quality and strong customer satisfaction.

The analysis also indicates that customer satisfaction is strongly associated with specialty food categories such as **desserts, ice creams, bakery products, and beverages**. Restaurants within these segments consistently achieve higher ratings and maintain strong customer engagement.

Furthermore, several restaurants have successfully sustained ratings above 4.7 despite receiving hundreds or thousands of customer reviews, making them valuable benchmarks for service quality, product consistency, and customer experience.

Q7: What Factors Appear to Influence Customer Satisfaction?

The analysis examined **delivery time, price level, and restaurant specialization** to identify observable patterns associated with higher restaurant ratings. Customer satisfaction was evaluated using the restaurant's average rating.

1. Delivery Time vs Ratings

Restaurants were grouped into Fast, Medium, and Slow delivery categories.

SQL Analysis

```
WITH DeliveryGroups AS
(
    SELECT
        CASE
            WHEN Delivery_time < 40 THEN 'Fast'
            WHEN Delivery_time BETWEEN 40 AND 60 THEN 'Medium'
            ELSE 'Slow'
        END AS Delivery_Category,
        Avg_ratings
    FROM Swiggy
)
SELECT
```

```

    Delivery_Category,
    COUNT(*) AS Restaurant_Count,
    ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM DeliveryGroups
GROUP BY Delivery_Category
ORDER BY
    CASE Delivery_Category
        WHEN 'Fast' THEN 1
        WHEN 'Medium' THEN 2
        WHEN 'Slow' THEN 3
    END;

```

Result

Delivery Category	Restaurant Count	Average Rating
Fast (<40 min)	1,634	3.82
Medium (40–60 min)	4,363	3.64
Slow (>60 min)	2,683	3.58

Restaurants with delivery times below 40 minutes recorded the highest average rating (**3.82**), while restaurants taking more than 60 minutes recorded the lowest (**3.58**). The **0.24-point difference** indicates an association between shorter delivery times and higher restaurant ratings.

2. Price Level vs Ratings

Restaurants were grouped into Low, Medium, and High price categories.

SQL Analysis

```

WITH PriceGroups AS
(
    SELECT
        CASE
            WHEN Price < 200 THEN 'Low Price'
            WHEN Price BETWEEN 200 AND 400 THEN 'Medium Price'
            ELSE 'High Price'
        END AS Price_Category,
        Avg_ratings
    FROM Swiggy
    WHERE Price > 0
)
SELECT
    Price_Category,
    COUNT(*) AS Restaurant_Count,
    ROUND(AVG(Avg_ratings), 2) AS Avg_Rating

```

```

FROM PriceGroups
GROUP BY Price_Category
ORDER BY
    CASE Price_Category
        WHEN 'Low Price' THEN 1
        WHEN 'Medium Price' THEN 2
        WHEN 'High Price' THEN 3
    EN

```

Result

Price Category	Restaurant Count	Average Rating
Low Price	977	3.66
Medium Price	5,911	3.61
High Price	1,787	3.79

High-priced restaurants recorded the highest average rating (**3.79**). However, the category comparison alone does not establish that higher prices cause higher ratings.

3. Restaurant Specialization

The analysis of highly rated restaurants showed that specialty categories such as **ice cream, desserts, bakery, and coffee** were frequently represented among top-performing restaurants.

This finding is based on the high-rated restaurant analysis presented earlier in the EDA and is therefore **not repeated here with another SQL query**.

The pattern indicates strong rating performance within these specialty categories, but does not establish that specialization itself causes higher ratings.

4. Correlation Analysis

To further examine the relationships between price, delivery time, and ratings, **Pearson correlation coefficients** were calculated using restaurant-level data.

SQL Analysis

```

WITH BaseData AS
(
    SELECT

```

```

        Price,
        Avg_ratings,
        Delivery_time
    FROM Swiggy
    WHERE Price > 0
),
Stats AS
(
    SELECT
        AVG(Price) AS Avg_Price,
        AVG(Avg_ratings) AS Avg_Rating,
        AVG(Delivery_time) AS Avg_Delivery
    FROM BaseData
)
SELECT
    'Price vs Rating' AS Relationship,
    CAST(
        SUM((b.Price - s.Avg_Price) *
            (b.Avg_ratings - s.Avg_Rating))
        /
        SQRT(
            SUM(POWER(b.Price - s.Avg_Price, 2)) *
            SUM(POWER(b.Avg_ratings - s.Avg_Rating, 2))
        )
        AS DECIMAL(10,4)
    ) AS Correlation
FROM BaseData b
CROSS JOIN Stats s

UNION ALL

SELECT
    'Delivery Time vs Rating',
    CAST(
        SUM((b.Delivery_time - s.Avg_Delivery) *
            (b.Avg_ratings - s.Avg_Rating))
        /
        SQRT(
            SUM(POWER(b.Delivery_time - s.Avg_Delivery, 2)) *
            SUM(POWER(b.Avg_ratings - s.Avg_Rating, 2))
        )
        AS DECIMAL(10,4)
    )
FROM BaseData b
CROSS JOIN Stats s

UNION ALL

SELECT
    'Price vs Delivery Time',
    CAST(
        SUM((b.Price - s.Avg_Price) *
            (b.Delivery_time - s.Avg_Delivery))
        /
        SQRT(
            SUM(POWER(b.Price - s.Avg_Price, 2)) *

```

```

        SUM(POWER(b.Delivery_time - s.Avg_Delivery, 2))
    )
    AS DECIMAL(10,4)
)
FROM BaseData b
CROSS JOIN Stats s;

```

Result

Relationship	Pearson Correlation
Price vs Rating	+0.1135
Delivery Time vs Rating	-0.1467
Price vs Delivery Time	+0.0755

Interpretation

The correlation results indicate **weak linear relationships** between the variables.

- **Price vs Rating (+0.1135):** weak positive association.
- **Delivery Time vs Rating (-0.1467):** weak negative association.
- **Price vs Delivery Time (+0.0755):** very weak positive association.

Therefore, while the category-level analysis shows differences in average ratings, **price and delivery time alone do not strongly explain variation in restaurant ratings across individual restaurants.**

5. Overall Finding

The analysis identified three observable patterns associated with restaurant ratings:

- **Faster delivery** — Fast-delivery restaurants recorded the highest average rating (**3.82**).
- **Price level** — High-priced restaurants recorded the highest average rating (**3.79**).
- **Restaurant specialization** — Specialty categories such as ice cream, desserts, bakery, and coee were frequently represented among highly rated restaurants.

However, the correlation analysis shows that the relationships between price, delivery time, and ratings are **weak**. Therefore, other factors such as food quality, service, cuisine, location, restaurant brand, and customer expectations may also contribute to differences in ratings.

Conclusion

The analysis suggests that delivery time, price level, and restaurant specialization are associated with differences in restaurant ratings, but the relationships are relatively weak. These findings should therefore be viewed as areas for further investigation rather than evidence of direct causation.

Advanced Analysis

1. City Level Performance Analysis

Objective

Evaluate the overall performance of each city by analyzing restaurant concentration, pricing, customer satisfaction, customer engagement, and delivery efficiency.

```
WITH CityPerformance AS
(
    SELECT
        City,
        COUNT(*) AS Total_Restaurants,
        ROUND(AVG(Price),2) AS Avg_Price,
        ROUND(AVG(Avg_ratings),2) AS Avg_Rating,
        SUM(Total_ratings) AS Total_Customer_Ratings,
        ROUND(AVG(Delivery_time),2) AS Avg_Delivery_Time
    FROM Swiggy
    WHERE Price > 0
    GROUP BY City
)

SELECT *,
    RANK() OVER
    (
        ORDER BY Avg_Rating DESC
    ) AS City_Rank
FROM CityPerformance;
```

Findings

- Chennai ranked **#1 in customer satisfaction** with an average rating of **3.78**, while maintaining a large restaurant base of **1,105 restaurants**.
- Kolkata had the **largest restaurant presence (1,346 restaurants)** but also recorded the **slowest average delivery time (67 minutes)**.
- **Kolkata** recorded an **average delivery time 52.3% higher** than **Ahmedabad**, indicating an area that may warrant further investigation into delivery operations.
- Mumbai emerged as the **most premium market**, with the highest average restaurant price (**₹393.79**), though higher pricing did not directly translate

into higher ratings.

- Hyderabad generated the **highest customer engagement**, recording **330,270 customer ratings**, which was **50.3% higher than Kolkata** despite having fewer restaurants.
- Delhi ranked last in customer satisfaction with an average rating of **3.53**, while Chennai outperformed Delhi by approximately **7.1%**.

2. Top-Performing Restaurants Within Each City

For this analysis, we selected restaurants with a minimum of 100 total ratings.

```
WITH RankedRestaurants AS
(
    SELECT
        City,
        Restaurant,
        Avg_ratings,
        Total_ratings,
        ROW_NUMBER() OVER
        (
            PARTITION BY City
            ORDER BY Avg_ratings DESC,
                    Total_ratings DESC
        ) AS rn
    FROM Swiggy
    WHERE Total_ratings >= 100
)

SELECT *
FROM RankedRestaurants
WHERE rn <= 5;
```

Findings

- **Nic Natural Ice Creams** emerged as a top performer in multiple cities, ranking #1 in Kolkata and Surat, demonstrating strong brand consistency across locations.
- **Corner House Ice Cream (Bangalore)** and **Mithai (Kolkata)** stood out with **1,000 customer ratings**, indicating sustained customer satisfaction at scale.

- Dessert, ice cream, bakery, and specialty food brands dominated the top rankings across almost every city.
- Chennai showed remarkable consistency, with all top 5 restaurants maintaining ratings of **4.7** and **500 customer ratings** each.
- Mumbai's highest-ranked restaurant, **Blue Tokai Coffee Roasters**, achieved a rating of **4.8**, highlighting strong customer preference for premium coffee and specialty food brands.
- Several top-performing restaurants belonged to well-established chains such as **Theobroma**, **Natural Ice Cream**, **Nic Natural Ice Creams**, and **Kwality Walls**, suggesting that brand reputation contributes positively to customer satisfaction.

3. Statistical Distribution

Price Distribution:

```
SELECT
    MIN(Price) AS Min_Price,
    MAX(Price) AS Max_Price,
    ROUND(AVG(Price),2) AS Avg_Price,
    ROUND(STDEV(Price),2) AS Std_Dev_Price
FROM Swiggy
WHERE Price > 0;
```

Findings:

- Restaurant prices ranged from **₹1 to ₹2,500**, indicating a wide variety of pricing options on the platform.
- The average restaurant price was **₹348.65**, while the standard deviation was **₹230.86**, showing significant variation in pricing.
- A small number of premium-priced restaurants exist, with the highest-priced restaurants charging over **7 times** the average price.
- Only **15 restaurants (0.17% of the dataset)** had prices below ₹10, suggesting potential anomalies or special pricing cases.

Rating Distribution:

```
SELECT
    MIN(Avg_ratings) AS Min_Rating,
    MAX(Avg_ratings) AS Max_Rating,
    ROUND(AVG(Avg_ratings),2) AS Avg_Rating,
    ROUND(STDEV(Avg_ratings),2) AS Std_Dev_Rating
FROM Swiggy;
```

Findings:

- Restaurant ratings ranged from **2.0 to 5.0**, indicating varying levels of customer satisfaction across the platform.
- The average restaurant rating was **3.66**, suggesting that most restaurants maintain moderate to good customer satisfaction.
- The standard deviation of **0.65** is relatively low compared to the rating scale, indicating that ratings are fairly concentrated around the average.
- Most restaurants appear to operate within a similar rating range, with relatively few extreme high- or low-performing restaurants.

Delivery Time Distribution:

```
SELECT
    MIN(Delivery_time) AS Min_Delivery,
    MAX(Delivery_time) AS Max_Delivery,
    ROUND(AVG(Delivery_time),2) AS Avg_Delivery,
    ROUND(STDEV(Delivery_time),2) AS Std_Dev_Delivery
FROM Swiggy;
```

Findings:

- Delivery times ranged from **20 to 109 minutes**, showing considerable variation in service speed across restaurants.
- The average delivery time was **53 minutes**.
- The standard deviation of **14.29 minutes** indicates moderate variability in delivery performance.
- The slowest deliveries took more than **5 times longer** than the fastest deliveries, highlighting operational differences across locations and restaurants.

Median:

Median Price:

```
SELECT DISTINCT
  PERCENTILE_CONT(0.5)
  WITHIN GROUP(ORDER BY Price)
  OVER() AS Median_Price
FROM Swiggy
WHERE Price > 0;
```

Median Rating:

```
SELECT DISTINCT
  PERCENTILE_CONT(0.5)
  WITHIN GROUP(ORDER BY Avg_ratings)
  OVER() AS Median_Rating
FROM Swiggy;
```

Median Delivery Time:

```
SELECT DISTINCT
  PERCENTILE_CONT(0.5)
  WITHIN GROUP(ORDER BY Delivery_time)
  OVER() AS Median_Delivery_Time
FROM Swiggy;
```

Price Distribution Analysis:

Findings:

- Restaurant prices vary significantly, ranging from **₹1 to ₹2500**.
- **Price is right-skewed** — the typical restaurant charges **₹300 (median)** but a small number of premium restaurants pull the average to **₹348.65**. This means if Swiggy uses average price in marketing communications, it overstates what most customers actually experience by **₹48.65** per order.

- The standard deviation (**₹230.86**) represents **66.2% of the average price**, highlighting substantial pricing variability across restaurants.
- Premium restaurants significantly influence the overall average price.

Customer Rating Distribution Analysis:

Findings:

- Customer ratings range from **2.0 to 5.0**, covering the full rating spectrum.
- The median rating (**3.90**) is **6.7% higher** than the average rating (**3.66**), suggesting that a relatively small number of poorly rated restaurants lower the platform average.
- The standard deviation (**0.65**) represents only **17.8% of the average rating**, indicating low variability in customer satisfaction.
- Most restaurants are clustered around ratings between 3.5 and 4.5.

Delivery Time Distribution Analysis:

Findings:

- Delivery times range from **20 to 109 minutes**, showing considerable operational variation.
- The average and median delivery times are identical (**53 minutes**), indicating a relatively balanced distribution.
- The standard deviation (**14.29 minutes**) accounts for **27.0% of the average delivery time**, suggesting moderate delivery variability.
- The slowest deliveries take **445% longer** than the fastest deliveries.

KPI Definition

KPI	Definition / Formula	Business Purpose
Total Restaurants	Count of all restaurants in the dataset	Measures the overall size of the restaurant marketplace
Average Rating	Average of Avg_ratings	Measures overall customer satisfaction
Average Delivery Time	Average of Delivery_time	Measures operational efficiency and delivery performance
Fast Delivery Percentage	Percentage of restaurants with delivery time less than 40 minutes	Measures delivery speed performance
High Rated Restaurants	Count of restaurants with rating greater than or equal to 4.5	Measures the number of top-performing restaurants
Total Cities	Count of distinct cities	Measures platform geographic coverage
Average Price	Average restaurant price excluding invalid values	Measures overall pricing level of restaurants on the platform

1. Customer Satisfaction KPIs

- **Average Rating**

SQL:

```
SELECT ROUND(AVG(Avg_ratings), 2) AS Avg_Rating
FROM   Swiggy;
```

Avg Rating : 3.66



The platform average rating is **3.66** out of **5 (73.2% score)**. With the highest-rated city (**Chennai**) at **3.78** and **lowest (Delhi)** at **3.53**, there is a **0.25**-point spread across cities — suggesting inconsistent customer experience depending on location.

- **High Rated Restaurants**

SQL:

```
SELECT COUNT(*) AS High_Rated_Restaurants
FROM Swiggy
WHERE Avg_ratings >= 4.5;
```

High Rated Restaurants : 662



Only **662** of **8,680** restaurants (**7.6%**) achieve a rating of **4.5** or above — meaning **92.4%** of the platform's restaurants fall below the high-performance threshold. This identifies a large opportunity for quality improvement across the majority of listed partners.

2. Operational KPIs:

- **Average Delivery Time**

SQL:

```
SELECT ROUND(AVG(Delivery_time), 2) AS Avg_Delivery_Time
FROM Swiggy;
```

Average Delivery Time: 53 min

- **Fast Delivery Percentage**

SQL:

```
SELECT Concat(CAST(COUNT(CASE WHEN Delivery_time < 40 THEN 1 END) * 100.0 / COUNT(*) AS
S DECIMAL(10, 2)), ' %') AS Fast_Delivery_Percentage
FROM Swiggy;
```

- **Percentage of restaurants providing fast delivery was : 18.82 %**

3. Marketplace KPIs:

- **Total Restaurants**

SQL:

```
SELECT COUNT(*) AS Total_Restaurants
FROM Swiggy;
```

Total Restaurants: 8680

- **Total Cities**

SQL:

```
SELECT COUNT(DISTINCT City) AS Total_Cities
FROM Swiggy;
```

Total Cities: 9

- **Average Price:**

SQL:

```
SELECT ROUND(AVG(Price), 2) AS Avg_Price
FROM Swiggy
WHERE Price > 0;
```

Average Price: 348.65/-



The platform average price is **₹348.65** — **sitting in the upper half of the medium-price band (₹200–400)**, just **13.4% below the ₹400 premium threshold**. This suggests most restaurants are priced near the premium boundary, with a natural upgrade opportunity for quality improvements.

Business Insights Summary

Insight 1: Kolkata Dominates Restaurant Availability but Faces Delivery Challenges

Kolkata has the largest restaurant base (1,346) but also the slowest average delivery time (67 minutes), highlighting a potential operational area for investigation.

Insight 2: Chennai Leads Customer Satisfaction

Chennai recorded the highest average restaurant rating at 3.78, making it the strongest-performing city by average rating in this dataset.

Insight 3: Faster Delivery Is Associated With Higher Ratings

Restaurants in the fast-delivery category (**<40 minutes**) recorded an average rating of **3.82**, compared with **3.58** for slow-delivery restaurants. The restaurant-level correlation between delivery time and rating was **-0.1467**, indicating a weak negative association rather than causation.

Insight 4: High-Priced Restaurants Have Higher Average Ratings

High-priced restaurants recorded the highest average rating (**3.79**), compared with **3.66** for low-priced and **3.61** for medium-priced restaurants. However, the price-rating correlation was only **+0.1135**, indicating a weak relationship rather than a strong pricing effect.

Insight 5: Hyderabad Demonstrates Exceptional Customer Engagement

Hyderabad generated **330,270** customer ratings, the highest among all cities in the dataset.

Despite having fewer restaurants than Kolkata, Hyderabad generated **50.3% more customer ratings**, indicating stronger customer engagement within the dataset.

Insight 6: Specialty Categories Are Frequently Represented Among Top-Rated Restaurants

Several highly rated restaurants belong to specialty categories such as **ice cream, desserts, bakery, and coffee**.

Restaurants including **Nic Natural Ice Creams, Natural Ice Cream, Theobroma, and Krispy Kreme** achieved ratings above **4.5**, suggesting strong customer satisfaction among several specialty-focused brands.

Insight 7: Nic Natural Ice Creams Emerged as a Strong Multi-City Performer

Nic Natural Ice Creams maintained an average rating of **4.59** across **9 cities** while accumulating more than **4,000 customer ratings**.

Its consistently high ratings across multiple markets indicate strong and broadly consistent customer satisfaction within the dataset.

Insight 8: Restaurant Pricing Shows Significant Variability

Restaurant prices ranged from ₹1 to ₹2500, with an average price of ₹348.65 and a median price of ₹300.

The average price was **16.2%** higher than the median, indicating a right-skewed pricing distribution where a relatively small number of premium restaurants increase the overall average.

Insight 9: Customer Ratings Show Less Variation Than Pricing

Customer ratings had an average of **3.66** with a standard deviation of **0.65**, indicating relatively limited variation compared with the much wider spread observed in restaurant prices.

Recommendations

Recommendation 1: Improve Delivery Efficiency in Kolkata

Based On

- Kolkata has the largest restaurant base (1,346).
- Kolkata has one of the slowest average delivery times (67–68 minutes).

Recommendation

Target: Reduce Kolkata's average delivery time toward the platform average of approximately 54 minutes. Swiggy should investigate delivery bottlenecks such as restaurant preparation time, rider availability, and delivery-area coverage before setting operational targets.

Recommendation 2: Learn From High-Performing Restaurant Categories

Based On

- Several highly rated restaurants belong to ice cream, dessert, bakery, and coffee categories.

Recommendation

Identify common characteristics among highly rated specialty restaurants and use these findings to support restaurant-partner benchmarking and category-level performance analysis.

Recommendation 3: Encourage Faster Deliveries

Based On

- Fast-delivery restaurants average 3.82 rating.
- Slow-delivery restaurants average 3.58 rating.

Recommendation

Investigate delivery-time bottlenecks and establish realistic delivery-performance benchmarks, particularly for cities and restaurant segments with consistently longer delivery times.

Recommendation 4: Learn From Top-Performing Restaurants

Based On

- Nic Natural Ice Creams
- Corner House Ice Cream
- Mithai
- Theobroma

achieved high customer ratings.

Recommendation

Analyze the operating and customer-experience characteristics of top-performing restaurants and identify practices that could be shared with lower-performing restaurant partners.

Recommendation 5: Replicate Chennai's Success

Based On

- Chennai recorded the highest average rating (3.78).

Recommendation

Analyze Chennai's restaurant mix, delivery performance, and customer-experience patterns to identify practices that may be transferable to lower-rated cities.

Recommendation 6: Focus on High-Engagement Markets

Based On

- Hyderabad generated 330,270 customer ratings, the highest among cities.
- This is 50.3% higher than Kolkata.

Recommendation

Use high-engagement markets such as Hyderabad as benchmarks for customer engagement and restaurant-partner performance. Investigate the factors contributing to higher review activity and identify practices that could be applied to other markets.

Limitations

This analysis provides valuable insights into restaurant performance; however, several limitations should be considered when interpreting the results.

- The dataset is sourced from Kaggle and may not fully represent the current Swiggy platform.
- The analysis is based on a single cross-sectional dataset and does not capture seasonal or time-based trends.
- Customer-level order history and transaction data were not available, limiting deeper behavioral analysis.
- The findings identify associations between variables (such as pricing, delivery time, and ratings) but do not establish causal relationships.
- Operational factors such as traffic conditions, restaurant preparation time, and delivery partner availability were not included in the dataset and may influence delivery performance.
- Customer ratings may be affected by factors beyond those available in the dataset, including food quality, customer expectations, promotions, and service experience.

Power Bi Dashboard

Executive Overview Dashboard

Purpose: Provides a high-level overview of the Swiggy restaurant ecosystem.

Highlights

- Total Restaurants
- Average Rating
- Average Delivery Time
- Average Price
- Fast Delivery %
- Restaurant Distribution by City
- Cuisine Analysis
- Delivery Performance
- Pricing Insights



Restaurant Performance Dashboard

Purpose: Evaluates restaurant performance, customer engagement, delivery efficiency, and pricing patterns.

Highlights

- Top-Rated Restaurant
- Fast Delivery Rating
- Top-Rated City
- Restaurants Rated 4.5+
- Top 10 Most Popular Restaurants
- Restaurant Performance Leaderboard
- Delivery Speed vs Customer Rating
- Average Rating by Price Category



PowerBI File

Swiggy_Dashboard.pbix